Primitive Roots for a Fixed Admissible Base: Exact Euler Corrections, Mellin Weights, and Conditional Analytic Refinements

Dr. Muralidhara Subbarao
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www.integralresearch.net
Stony Brook University
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Abstract

We study the theta–reflection–Poisson interface in the supplied manuscript *Primitive roots for every admissible integer base*. We preserve its fixed-base quantifiers and distinguish arithmetic inputs from consequences of integral transforms. At the principal Poisson residue we factor the exact local correction, obtain nonvanishing for real part greater than 2/3, and isolate a residual product holomorphic and nonzero for real part greater than 1/2. We construct annular weights with arbitrarily prescribed finite-order Fourier-origin cancellation while preserving the nonzero principal Mellin residue. Conditional on the stated conductor large sieve and growth bounds, a shorter zero detector improves the exceptional-row exponent. A separate exact contour ledger retains the reflection-limited exclusion width 1/6000 and reduces the required far Mellin line from 350000 to 4000; we also characterize the least integer line for a restricted two-inequality design. These are local algebraic and conditional analytic refinements, not verification of the source's global zero-free theorem, Kummer uniformity, marked Type II estimate, sieve construction, or fixed-base Artin claim. Contribution assessment: C3/5, notable explicit local refinements and conditional analytic savings, with the global arithmetic inputs unproved here.

1 Scope, Quantifiers, and Dependencies

The public source is the October 4, 2026 manuscript in the pinned local source tree [1]. Its main claim is the following statement: for each fixed integer a that is neither -1 nor an integer square, there exist $c_a > 0$ and x_a such that every real $x \geq x_a$ has at least $c_a x/(\log x)^2$ primes $x < p < 2x$ with $p \nmid a$ and $\mathrm{ord}_p(a) = p - 1$. The admissibility condition excludes 0 and 1, so $|a| > 1$. This is not an asserted density asymptotic, and its constants are not uniform in a. In this report it remains a *source claim*.

The source also claims a common all-height, all-conductor Hecke zero-free width 10^{-6} over each fixed cyclotomic field containing μ_{12}. Its proof actually obtains width 1/20000 before weakening the numerical statement. Fixed field and target character are chosen before the auxiliary data; constants and starting points may depend on them. Reading this argument is not certification of this very strong analytic input.

Source component	Role and retained boundary
Arithmetic; coefficient family	Self-dual S-integer lattice, allowed generators, sextic masks, phase corrections, and the precise smoothed coefficient.
Theta; Whittaker appendix	Exceptional representation, rational and compact lifts, normalized local coefficients, global factorization and its correction. These automorphic inputs are imported.
Reflection	Weyl calculation produces the quadratic pairing $1-4\equiv 3$ modulo 6; a quadratic Hecke large sieve bounds the reflected moment.
Poisson; zero-free	Finite Fourier transform, frequency decomposition, Euler factors, zero detection, contour separation and Mellin inversion. The local improvements below are concentrated here.
Splitting; Type II	Growing Kummer fields and marked bilinear prime correlations require independent arithmetic uniformity.
Sieve; construction; reduction	Produce $p-1=crQ$, remove possible prime divisors of the index, and retain the fixed-base quantifiers.

We use three status categories throughout: *local* means proved below from explicitly given finite identities or analytic hypotheses; *conditional* means a deduction from named source estimates; *imported* means not independently proved in this report. No inference from a publisher, a source snapshot, or a positive Gram matrix substitutes for a missing arithmetic estimate. The read-scope and proof-boundary evidence accompanying this report identify actual literal reads and unread complements.

2 Measures, Phases, and the Ram Interface

Fix a cyclotomic field $F\supset\mu_{12}$ of degree $2r$, a finite-order Hecke character η, and the source's final fixed set S. Set $R=\mathcal{O}_S$. Exterior ideals are supported away from S and $q_{\mathfrak{a}}=\mathrm{N}\,\mathfrak{a}$. The finite norm is $|\varpi_v|_v=(\mathrm{N}\,v)^{-1}$; at a complex place it is the *square* of the ordinary modulus. Write $F_S=\prod_{v\in S}F_v$ and $|y|_S=\prod_v|y_v|_v$. The source uses the trace character and self-dual additive measure, with

$$\widehat{W}(y)=\int_{F_S}W(x)\mathrm{e}_S(-xy)\,\mathrm{d}x,\qquad \mathrm{covol}(R)=1.$$

Thus Poisson on an exterior ideal bR has factor q_b^{-1}, not $q_b^{-1/2}$. Additive dilation by y has modulus $|y|_S$. Multiplicative Haar measure instead satisfies

$$\int_{F_S^\times}f(y)\,\mathrm{d}^\times y=\int_0^\infty\int_{|y|_S=T}f(y)\,\mathrm{d}_T^\times y\,\frac{\mathrm{d}T}{T}. \tag{1}$$

Multiplication and inversion on norm sheets introduce no extra norm Jacobian. Haar constants are fixed once, and cannot subsequently be hidden in a claimed principal coefficient without being recorded.

The original private master-atoms chapter proves, in its Mellin section, that for $g_c(t)=\mathrm{e}^{ct}f(\mathrm{e}^t)\in L^1(\mathbb{R})\cap L^2(\mathbb{R})$,

$$\mathcal{M}f(c+i\tau)=\widehat{g}_c(-\tau),\qquad \int_0^\infty|f(x)|^2x^{2c}\frac{\mathrm{d}x}{x}=\frac{1}{2\pi}\int_{\mathbb{R}}|\mathcal{M}f(c+i\tau)|^2\,\mathrm{d}\tau, \tag{2}$$

where $\mathcal{M}f(s)=\int f(x)x^s\,\mathrm{d}x/x$ and the Fourier kernel on $\mathbb{R}$ is $\mathrm{e}^{-it\tau}$. Substituting $x=\mathrm{e}^t$ proves both identities; there is no additional e^t Jacobian after using $\mathrm{d}x/x$. This is an exact weighted

coordinate change, not a cancellation estimate. The private TR1674 proof gives $K = C^*WC$ for a finite weighted prefix operator and explains why $K \succeq 0$ alone does not bound its arithmetic input. Here the analogous quadratic sieve is an imported bound on a particular family, not a consequence of Gram positivity. Complex coefficients require the adjoint C^*, not the unconjugated transpose.

In the source, every positive-modulus character power is zero on nonunits, *including powers divisible by six*. For allowed exterior generators, the unitary Hecke character $\vartheta = \overline{\Lambda}\,\overline{G}\eta$ satisfies

$$\gamma_1(\mathfrak{p})\gamma_2(\mathfrak{p})\vartheta(\mathfrak{p}) = -\eta(\mathfrak{p}). \tag{3}$$

This negative sign determines the reciprocal L-factor. It is not recoverable from absolute values or an energy calculation. The coefficient family is

$$B_m(Z) = \sum_{\substack{\mathfrak{c}\ \text{squarefree} \\ \mathfrak{n}}} \frac{\gamma_2(\mathfrak{c})\vartheta(\mathfrak{c}\mathfrak{n}^3)\chi_{\mathfrak{c}\mathfrak{n}^3}(m)}{q_{\mathfrak{c}}^{1/2} q_{\mathfrak{n}}} P_G(q_{\mathfrak{c}\mathfrak{n}^3}/Z), \quad P_G(t) = \frac{\mathrm{e}^{-(\log t)^2/4}}{2\sqrt{\pi}}. \tag{4}$$

The two ideals may overlap. Its absolute Mellin majorant for real $\sigma > 1/2$ is $\zeta_F^S(\sigma + 1/2)\zeta_F^S(3\sigma + 1)$, and $\mathcal{M}P_G(s) = \mathrm{e}^{s^2}$. The mask gives $B_0(Z) = P_G(1/Z)$.

3 An Exact Principal Euler Correction

This section is a local algebraic refinement of the actual good-prime calculation in `05-poisson.tex`, not a replacement for that calculation's Gauss sums or CRT phases. At the principal frequency $u = 1$ and the double residue $w = 1, z = 1/6$, let $Q = q_{\mathfrak{p}}$ and

$$r_Q = Q^{-1}, \qquad D = \eta(\mathfrak{p})Q^{-s}, \qquad E = \vartheta(\mathfrak{p})^6 Q^{3-6s}.$$

The source's exact rational factor becomes

$$F_{\mathfrak{p}} = \frac{1}{1 - r_Q} + \frac{r_Q}{1 - r_Q} + \frac{E}{1 - E}(1 + r_Q) - \frac{D}{1 - E}(1 + r_Q), \tag{5}$$

$$H_{\mathfrak{p}} = \frac{(1 - r_Q)^2}{1 - D} F_{\mathfrak{p}}. \tag{6}$$

Theorem 3.1 (Factorization and domains)**.** *For* $\operatorname{Re} s > 1/2$*, each displayed local factor is defined and nonzero, and*

$$H_{\mathfrak{p}}(s) = \frac{1 - Q^{-2}}{1 - E} J_{\mathfrak{p}}(s), \quad J_{\mathfrak{p}}(s) = 1 + \frac{Q^{-1}(D - E)}{1 - D}. \tag{7}$$

The product $J_\eta(s) = \prod_{\mathfrak{p} \notin S} J_{\mathfrak{p}}(s)$ *is holomorphic and nowhere zero on* $\operatorname{Re} s > 1/2$*. On every closed half-plane* $\operatorname{Re} s \geq 1/2 + \epsilon$ *it and its reciprocal are bounded, with constants depending on* F, S, ϵ*. For* $\operatorname{Re} s > 2/3$ *the principal correction converges absolutely, is nonzero, and equals*

$$H_\eta(s) = \frac{L_F^S(6s - 3, \vartheta^6)}{\zeta_F^S(2)} J_\eta(s). \tag{8}$$

No additional small exterior primes need be removed for this principal nonvanishing statement.

Proof. Multiplying (5) by the normalizing factor gives

$$H_{\mathfrak{p}} = (1 - r_Q^2)\frac{1 - (1 - r_Q)D - r_Q E}{(1 - D)(1 - E)}.$$

This proves (7) algebraically, with arbitrary complex phases. Write $\sigma = \operatorname{Re} s > 1/2$. Unitarity gives $|D| = Q^{-\sigma} < 1$ and $|E| = Q^{3-6\sigma} < 1$. Consequently

$$|(1 - r_Q)D + r_Q E| \le (1 - r_Q)Q^{-\sigma} + r_Q Q^{3-6\sigma} < 1.$$

The numerator cannot vanish. Uniformly on $\sigma \ge \sigma_0 > 1/2$,

$$|J_{\mathfrak{p}} - 1| \le \frac{Q^{-1-\sigma} + Q^{2-6\sigma}}{1 - Q^{-\sigma}} \le C_{\sigma_0}(Q^{-1-\sigma_0} + Q^{-(6\sigma_0-2)}). \tag{9}$$

Both exponents exceed one. Prime-ideal summability, which follows from $\sum_{\mathfrak{a}} (\mathrm{N}\,\mathfrak{a})^{-b} < \infty$ for $b > 1$, proves normal convergence of the product. Its tail has an absolutely convergent logarithm; the finitely many initial factors and their reciprocals are uniformly bounded by the strict displayed inequality. This also proves nonvanishing and the claimed half-plane bounds.

For $\sigma > 2/3$, $\sum |E| < \infty$ as $6\sigma - 3 > 1$. Multiplying (7) gives (8) by absolutely convergent Euler products. The source's principal correction agrees here with this product on their common initial domain, so uniqueness of analytic continuation gives the same function wherever both are defined. □

The source proves principal nonvanishing only in the narrower region $\operatorname{Re} s \ge 0.998$ needed by its contours. Theorem 3.1 explains it explicitly and widens that local region. It does *not* assert that H_η is zero-free throughout $\operatorname{Re} s > 1/2$: after continuation, the Hecke factor in (8) may contribute zeros. Nor does it remove the source's exclusions at S required by the automorphic data, conductors, generators, or nonprincipal Euler factors.

4 Annular Weights with Controlled Origin Vanishing

The probe requires $W_0 \in \mathcal{S}(F_S)$ supported in $F_S^\times$, $\widehat{W_0}(0) = 0$, and

$$M(s) = \int_{F_S^\times} \widehat{W_0}(y)|y|_S^s \, \mathrm{d}^\times y, \qquad M(1/6) \ne 0.$$

The first support condition removes the original row $m = 0$; the Fourier condition removes the Poisson frequency $h = 0$. These are different requirements. The source constructs two annular bumps at one complex place and obtains M holomorphic for $\operatorname{Re} s > 0$. The next construction gives a controlled leftward extension of the *weight transform*, without changing any arithmetic Euler region.

Theorem 4.1 (Higher-order annular construction)**.** *For every fixed integer $J \ge 1$ one can choose such a tensor-product W_0 so that M is holomorphic on $\operatorname{Re} s > -J$ and rapidly decreasing vertically on every closed bounded real strip in that half-plane. Its norm-sheet profile $V_0(T) = \int_{|y|_S = T} \widehat{W_0}(y) \, \mathrm{d}_T^\times y$ satisfies, for $0 < \epsilon < J$ and $A > 0$,*

$$|(T\partial_T)^b V_0(T)| \le C_{J,\epsilon,A,b} \min(T^{J-\epsilon}, T^{-A}), \qquad T > 0. \tag{10}$$

All choices and constants are fixed before Z, U grow. No positivity of W_0 is asserted.

Proof. At a complex place choose a nonnegative nonzero smooth radial bump h in an annulus and $q > 1$. Let $|\alpha|_v = q$ and define the additive-mass-preserving dilation

$$(U_q h)(x) = q^{-1} h(x/\alpha), \qquad W_{v,J} = \prod_{j=0}^{J-1} (1 - q^{-j} U_q) h.$$

Every term is supported in an annulus away from zero. Since $\int |x|_v^k U_q h(x)\,\mathrm{d}x = q^k \int |x|_v^k h(x)\,\mathrm{d}x$,

$$\int |x|_v^k W_{v,J}(x)\,\mathrm{d}x = \prod_{j=0}^{J-1}(1-q^{k-j}) \int |x|_v^k h(x)\,\mathrm{d}x = 0 \quad (0 \le k < J). \tag{11}$$

The Fourier transform of a smooth radial function in two real dimensions has a Taylor expansion in $|y|^2_{\text{ordinary}} = |y|_v$. Angular integration kills odd and unbalanced monomials; the coefficient of $|y|_v^k$ is a fixed scalar times its kth radial moment. Taylor's remainder for the compactly supported bump therefore proves $\widehat{W}_{v,J}(y) = O(|y|_v^J)$, also after any fixed logarithmic radial derivative. At infinity it is Schwartz.

Fourier scaling gives $\widehat{U_q h}(y) = \widehat{h}(\alpha y)$, and multiplicative substitution gives, initially for $\operatorname{Re} s > 0$,

$$M_{v,J}(s) = \prod_{j=0}^{J-1}(1-q^{-j-s}) M_{v,h}(s). \tag{12}$$

The local Fourier–Mellin identity used in the source says, for $0 < s < 1$,

$$M_{v,h}(s) = c_v(s) \int_{F_v^\times} h(x)|x|_v^{1-s}\,\mathrm{d}^\times x, \qquad c_v(s) \neq 0.$$

Its normalization is with the above self-dual additive character and fixed multiplicative Haar measure. It follows from the Fourier transform of the homogeneous distribution $|x|_v^{s-1}$ (equivalently from Gaussian integration at a complex place). We use this standard local identity, not a global L-function theorem. The integral is positive for our h, so (12) is nonzero at $s = 1/6$.

At each finite place of S, choose a nonnegative nonzero locally constant annular function h_v. Take α with $|\alpha|_v = q > 1$ and set $W_v = (1 - U_q)h_v$. Its additive integral is zero, hence its locally constant Fourier transform vanishes on a neighborhood of zero. That Fourier transform is also compactly supported. Its Mellin transform is entire, and the same local identity and multiplier $1 - q^{-s}$ show nonvanishing at $1/6$. Rescaling a character of nonzero conductor only changes the fixed nonzero scalar in that identity.

Perform the complex construction at *every* complex place. Making it at only one place would not remove possible local obstructions at the others. Tensor the resulting factors. At each complex place, weighted absolute multiplicative integrability holds for $\operatorname{Re} s > -J$; at each finite place only finitely many Fourier valuation shells remain. Fubini therefore gives the product formula for M and its holomorphy. Logarithmic integration by parts at one complex place, with the endpoint bounds just proved, gives arbitrary vertical decay on closed strips.

For completeness, parametrize a norm sheet by all local variables except the norm t_0 at one complex place. Then $t_0 = T\prod_{v \neq v_0} |y_v|_v^{-1}$. For any $\sigma > -J$, take the weighted supremum $|y_{v_0}|^\sigma |\mathcal{D}_{v_0}^b \widehat{W}_{v_0}|$ and the weighted absolute multiplicative integrals at the other places. Their product bounds $T^\sigma |(T\partial_T)^b V_0(T)|$. Taking $\sigma = -J + \epsilon$ and $\sigma = A$ proves (10). □

This construction has an explicit cost: $\|W_{v,J}\|_{L^1(dx)} \le \prod_{j<J}(1+q^{-j})\|h\|_1$. Its support grows through finitely many dilated annuli, and its higher seminorms and residue normalization need not be uniform in J. Thus J cannot silently grow with the conductor. The theorem improves controllable weight tails and supplies reproducible nonzero residues; it does not justify shifting an arithmetic Euler product to $\operatorname{Re} z \le 0$.

5 A Shorter Conditional Zero Detector

We now isolate assumptions from `06-zero-free.tex`, so the analytic saving is not mistaken for a new unconditional large sieve. Fix F, S, η. For a dyad $U \le q_u < 2U$, use the distinct primitive inducing characters ω_u of $\eta\overline{\chi_\bullet(u)}$, of conductor $O_{F,S}(U)$. The source's Kummer distinctness is an input. Put $\sigma_0 = 249/250$. We assume:

(LS) For each $\epsilon > 0$ and any ideal coefficients of length N,

$$\sum_{\omega} \left| \sum_{\mathrm{N}\,\mathfrak{n} \le N} b_{\mathfrak{n}} \omega(\mathfrak{n}) \right|^2 \ll_{F,S,\epsilon} (N + U^2)(NU)^{\epsilon} \sum |b_{\mathfrak{n}}|^2.$$

The sum includes the family in question, with its specified finite local types; the assertion is not derived from finite Gram positivity.

(G) The nonprincipal Hecke functions are entire, admit the usual Mellin contour operations and polynomial strip growth, and obey $|L_F(1/2 + it, \omega)| \ll_\epsilon U^{1/2+\epsilon}(2 + |t|)^{A_F}$ for a fixed $A_F \ge 0$. Their Euler series converge absolutely to the right of one. The principal character is separated as at most one row.

(I) Fixed-field ideal counting and divisor estimates give $\sum_{\mathrm{N}\,\mathfrak{n} \le N} \tau_F(\mathfrak{n})^2 \ll_\epsilon N^{1+\epsilon}$. These imply all coefficient square-mass estimates used below.

Theorem 5.1 (Detector tradeoff)**.** *Let $\kappa > 2$, $\rho > 0$ be fixed and satisfy*

$$\kappa(\sigma_0 - 1/2) > 3/2 + A_F\rho. \tag{13}$$

Under (LS), (G), (I), the number of nonprincipal rows whose L-function has a zero $\beta + i\gamma$ with $\beta \ge \sigma_0$ and $|\gamma| \le 4U^\rho$ is, for every $\epsilon > 0$,

$$\ll_\epsilon U^{2\kappa(1-\sigma_0)+\rho+\epsilon}. \tag{14}$$

In particular $\kappa = 25/8$ and $0 < \rho < 1/[100(A_F + 1)]$ give $O_\epsilon(U^{1/40+\rho+\epsilon})$, and hence $O(U^{1/25})$ after fixing a sufficiently small loss. This improves the source's $O(U^{1/2})$ exceptional-row bound in this same conditional framework.

Proof. Let

$$D_U(s, \omega) = \sum_{\mathrm{N}\,\mathfrak{d} \le U^2} \frac{\mu_F(\mathfrak{d})\omega(\mathfrak{d})}{(\mathrm{N}\,\mathfrak{d})^s}, \qquad a_U(\mathfrak{n}) = \sum_{\substack{\mathfrak{d} \mid \mathfrak{n} \\ \mathrm{N}\,\mathfrak{d} \le U^2}} \mu_F(\mathfrak{d}).$$

Then $a_U(1) = 1$, $a_U(\mathfrak{n}) = 0$ for $1 < \mathrm{N}\,\mathfrak{n} \le U^2$, and $|a_U| \le \tau_F$. At a selected zero $z_0 = \beta + i\gamma$ use the exponential cutoff $Y = U^\kappa$. Mellin inversion gives

$$\sum_{\mathfrak{n}} a_U(\mathfrak{n})\omega(\mathfrak{n})(\mathrm{N}\,\mathfrak{n})^{-z_0} \mathrm{e}^{-\mathrm{N}\,\mathfrak{n}/Y} = \frac{1}{2\pi i} \int_{(2)} L_F(z_0 + v, \omega) D_U(z_0 + v, \omega) \Gamma(v) Y^v \, \mathrm{d}v.$$

Shift to $\mathrm{Re}\, v = 1/2 - \beta$. The apparent residue of Γ at zero vanishes because $L_F(z_0, \omega) = 0$; no principal pole is present. No further gamma pole is crossed. By (I), $D_U(1/2 + it, \omega) \ll_\epsilon U^{1+\epsilon}$. Gamma decay and (G) bound the new integral by

$$U^{\kappa(1/2-\beta)+3/2+A_F\rho+\epsilon} = o(1)$$

after choosing the preliminary loss below the strict gap in (13). The term at the unit ideal tends to one. The remaining sum, over norms greater than U^2, therefore has modulus bounded below by an absolute positive constant.

Truncate at $Y(\log U)^2$. The omitted tail is bounded by a fixed power of U times $\exp(-\frac{1}{2}(\log U)^2)$, using (I) and partial summation; it is smaller than every prescribed inverse power. There are $O(\log U)$ dyads left, so one dyadic sum has size $\gg 1/\log U$. On that dyad, $t^{\sigma_0-\beta}\mathrm{e}^{-t/Y}$ is decreasing and at most one. Partial summation reduces its lower bound to a maximum of prefixes of

$$P_{\omega,N}(v,t)=\sum_{N<\mathrm{N}\,\mathfrak{n}\le v} a_U(\mathfrak{n})\omega(\mathfrak{n})(\mathrm{N}\,\mathfrak{n})^{-\sigma_0-it},\quad N\le v\le 2N,$$

with $U^2/2<N\ll U^{\kappa}(\log U)^2$. Group equal norms together. Every prefix is a union of at most $O(\log N)$ disjoint binary intervals. Cauchy bounds its squared modulus by $O(\log N)$ times the sum of the squared interval polynomials. At each binary level, the coefficient square masses of those intervals add to the mass of the dyad; (LS) can thus be summed over all intervals with only a further logarithmic factor, not their number.

For a differentiable polynomial P on a unit height interval, choose a point where $|P|^2$ is at most its integral. Integrating $(|P|^2)'=2\,\mathrm{Re}(P'\overline{P})$ then proves

$$\sup_I |P|^2\le C\int_I (|P|^2+|P'|^2)\,\mathrm{d}t.$$

Cover $[-4U^{\rho},4U^{\rho}]$ by unit intervals. At every height apply (LS) to each binary interval; differentiation multiplies coefficients by $-i\log\mathrm{N}\,\mathfrak{n}$, costing only logarithms. By (I), the coefficient mass is $\ll_\epsilon N^{1-2\sigma_0+\epsilon}$. Since $N>U^2/2$, the resulting height-integrated bound is

$$\ll_\epsilon U^{\rho+\epsilon}N^{2-2\sigma_0+\epsilon}\ll_\epsilon U^{\rho+2\kappa(1-\sigma_0)+\epsilon}.$$

Absorb the dyad, prefix and detection logarithms by renaming the arbitrary loss. Selecting one zero per exceptional row proves (14); zeros are not being counted with multiplicity. Finally $(25/8)(\sigma_0-1/2)-3/2=1/20$ and $2(25/8)(1-\sigma_0)=1/40$. The stipulated $\rho<1/100$ leaves room to choose the loss so that the count is $O(U^{1/25})$. □

The source used $Y=U^{10}$ and a cutoff U^{11}, giving before slack $U^{11(2-2\sigma_0)+\rho+\epsilon}=U^{0.088+\rho+\epsilon}$. The new truncation retains the exponential tail instead of paying a whole extra power of U. Constants and the allowed height exponent still depend on F. The reciprocal bound on good rows must still be proved from the zero-free rectangles, Euler control, and the three-circles argument; it is not supplied merely by (14). We use the source's version of that bound in the next section, and do not need this improved exceptional count for the stated contour certificate.

6 Contour Savings and the Mellin Endpoint

We state precisely which source interfaces are retained. Let

$$a_0=9/10,\quad b_0=133/1000,\quad M_0=a_0+b_0=1033/1000,\quad h_0=1-a_0=1/10,$$

and put

$$E_*(\sigma,w,z)=a_0/2+\sigma-1+h_0z+b_0(w-1),\quad C_*(s)=s-8/15,\quad C_*(1)=7/15.\tag{15}$$

Here a_0 is a scale exponent, not the integer primitive-root base a.

Imported analytic package (A). The exact Poisson identity, with the measures already fixed, has dyadic frequency rows u, a principal double residue $c_0 f_\eta$, and

$$\begin{aligned} f_\eta(Z) &= \frac{1}{2\pi i}\int_{(2)} Z^{s-8/15}\mathrm{e}^{(s-5/6)^2}\frac{H_\eta(s)}{L_F^S(s,\eta)}\,\mathrm{d}s, \\ c_0 &= \frac{(\mathrm{Res}_{s=1}\,\zeta_F^S(s))^2}{6}M(1/6)\widehat{W}_1(1)\neq 0. \end{aligned} \tag{16}$$

For fixed nonnegative nonzero $W_1\in C_c^\infty((0,\infty))$, $\widehat{W}_1(1)=\int W_1(t)\,\mathrm{d}t$. The factor $1/6$ is the residue Jacobian of $\zeta_F^S(6z)$. Source Euler estimates hold in neighborhoods of the two regions

$$(\mathrm{Re}\,\xi,\mathrm{Re}\,w,\mathrm{Re}\,z)\geq(.998,.003,.4),\qquad (\mathrm{Re}\,\xi,\mathrm{Re}\,w,\mathrm{Re}\,z)\geq(.998,.95,.16),$$

with $H_u\ll_\epsilon q_u^\epsilon$. Numerator Hecke functions obey the source's conservative $U^{1/2+\epsilon}$ bound for $.003\leq \mathrm{Re}\,w\leq 4$ and polynomial height growth. There are $O(U)$ rows per dyad and $O(U^{1/2})$ exceptional rows with zeros in $\mathrm{Re}\,s\geq .996$, $|\Im s|\leq 4U^\rho$, for some fixed $\rho>0$. On the other rows the source's reciprocal estimate is $1/L^S\ll_\epsilon U^\epsilon$ in the rectangle needed to move $\mathrm{Re}\,\xi$ to .998, $|\Im\xi|\leq U^\rho$. Outside it we use only absolute convergence on $\mathrm{Re}\,\xi>1$. These statements include their uniformity over u and the finite local types; they are assumptions here.

The height kernel is bounded on fixed real strips by

$$C_A\mathrm{e}^{-(\Im\xi+\Im z)^2}(1+|\Im z|)^{-A}(1+|\Im w|)^{-A}. \tag{17}$$

The Gaussian is in the *sum* of two heights. One may not replace it by separate Gaussian factors. Splitting the z-integral according to $|\Im z|\geq|\Im\xi|/2$ or its complement proves arbitrary polynomial decay in $\Im\xi$ after integration, which is sufficient for the tails.

Imported reflection package (R). The actual theta, Weyl and quadratic large-sieve argument gives, for every $\epsilon>0$ and the fixed compact shape sets used by the probe,

$$\sum_m |B_m(Z)|^2\ll_\epsilon Z^{2M_0-1+\epsilon}.$$

The additive probe has squared mass $O(Y^{-1})$, $Y=Z^{b_0}$. Cauchy, with a renamed loss, therefore gives

$$|I(Z^{a_0},Z^{b_0},Z)|\ll_\epsilon Z^{(2M_0-1-b_0)/2+\epsilon}=Z^{C_*(1)-1/6000+\epsilon}. \tag{18}$$

Both packages require the specified theta normalizations and arithmetic families. The local transform results above do not prove them.

Proposition 6.1 (A smaller fixed far line)**.** *Assume (A). Retain the source contours except that the intermediate range ends at $U=Z^t$ with $t=1009/10000$, and the far line is $\mathrm{Re}\,z=4000$. Then, for every $\epsilon>0$, the total nonprincipal contribution is*

$$O_\epsilon(Z^{C_*(1)-833/3000000+\epsilon}). \tag{19}$$

In particular, after choosing the loss, the full comparison, including the principal contour remainders, is

$$I(Z^{a_0},Z^{b_0},Z)=c_0f_\eta(Z)+O(Z^{C_*(1)-1/5000}). \tag{20}$$

The fixed far line is 87.5 times smaller than the source's 350000. This is a reduction in required weight-strip constants, not a uniform numerical bound on them.

Proof. Put $d_1 = 10^{-7}$, $\sigma_1 = 1 + d_1$. On the source's low-w contour $(\sigma_1, .003, .4)$, a subset of $O(U^{\alpha})$ rows is bounded by

$$Z^{E_*(\sigma_1,.003,.4)}U^{\alpha+.1+\epsilon}.$$

The relative Z exponent is $d_1 + 7/300 - .132601$. Small rows $U \le Z^{.09}$ use $\alpha = 1$; exceptional intermediate rows use $\alpha = 1/2$. For good rows $Z^{.09} < U \le Z^t$, move the finite ξ rectangle to .998. Its central contribution has relative exponent

$$-.002 + 7/300 - .132601 + 1.1t = -833/3000000. \tag{21}$$

All moves stay in the first Euler region and cross no numerator poles: these rows have $u \neq 1$. The reciprocal is used only within its permitted rectangle. Equation (17), after absorbing polynomial growth, bounds the original tails and horizontal segments by

$$Z^{E_*(\sigma_1,.003,.4)}U^{1.1+\epsilon-\rho A_1}.$$

Choose $A_1 > 2/\rho$ after fixing the field-dependent ρ. Since $U > Z^{.09}$, these terms are smaller than the central target. Increasing derivative orders is permitted for the fixed smooth weights.

For $U > Z^t$, keep $\operatorname{Re}\xi = \operatorname{Re} w = 4$ and move z right to $R = 4000$. This crosses no pole, remains in the first Euler region and uses only absolute Euler convergence for both Hecke factors. The bound is

$$Z^{3.849+.1R}U^{1-R+\epsilon}.$$

The dyadic tail converges geometrically. Evaluated at its lower endpoint, its absolute Z exponent is $3.849 + t - (t - .1)R = .3499$, safely below $C_*(1) - 1/6000$. The relevant margins, before arbitrary small losses, are as follows.

Contribution	Exponent relative to 7/15
Principal w remainder	$-.00664989$
Principal z remainder	$-1/1500 + 10^{-7}$
Small rows	$10^{-7} + 7/300 - .132601 + .099$
Exceptional rows to $Z^{.1009}$	$10^{-7} + 7/300 - .132601 + .06054$
Good central rows	$-833/3000000$
Far rows, $R = 4000$	$.3499 - 7/15$

The principal bounds follow by the same residues at $w = 1, z = 1/6$ and remaining lines $(\sigma_1, .95, 1/6 + d_1)$ and $(\sigma_1, 1, .16)$. Each margin other than the good central one is strictly larger in absolute value. The finitely ranged dyads cost $O(\log Z)$, absorbed by an arbitrary loss, proving (19). Since $833/3000000 > 1/5000 > 1/6000$, a sufficiently small loss proves (20). In particular the source's 0.0004 remainder cannot be carried unchanged to these modified contours. □

Proposition 6.2 (Restricted least integer line)**.** *For the unchanged small, principal and exceptional contours just used, the one-threshold design preserves a strict pre-loss margin beyond* $\Delta = 1/6000$ *precisely when its good and far bounds permit*

$$\frac{1}{10} + \frac{1393}{400(R-1)} < t < \frac{111101}{1100000}. \tag{22}$$

Among integer R, the least feasible value is 3481.

Proof. The good bound requires $.002-7/300+.132601-1.1t > 1/6000$, which is the upper inequality. The far bound requires $3.949-(R-1)(t-.1) < 7/15-1/6000 = .4665$, giving the lower one. Their interval is nonempty if and only if $R-1 > 3830750/1101$. Since $3479 < 3830750/1101 < 3480$, the least integer R is 3481. Choose the midpoint of the two rational endpoints at that R. It lies above .1 and below .101001, so the small and exceptional comparisons and the same tail argument retain ample slack. The good and far slack is positive but small, so the arbitrary preliminary losses must be chosen accordingly. This proves sufficiency as well as necessity for these two inequalities only. □

This does not optimize other contour coordinates, the reflection input, or global uniform constants. The simpler $R = 4000$ certificate is preferable when generous numerical slack matters.

Theorem 6.3 (Reflection-limited conditional exclusion)**.** *Assume (A), (R), and the usual Hecke meromorphic continuation. Then*

$$L_F(s,\eta) \neq 0 \qquad \left(\operatorname{Re} s > 1 - \frac{1}{6000}\right), \tag{23}$$

with the principal pole at $s = 1$ understood as a pole, not a zero. There is no conclusion on the boundary line. Constants and onsets need not be uniform as an auxiliary exponent approaches its endpoint.

Proof. Combine (18) with (20) and $c_0 \neq 0$. For every $\epsilon > 0$, $f_\eta(Z) \ll_\epsilon Z^{7/15-1/6000+\epsilon}$ for large Z. For $0 < Z \leq 1$, moving the defining contour in (16) rightwards gives $f_\eta(Z) = O_A(Z^A)$ for every fixed $A > 0$. This uses absolute Euler convergence and Gaussian height decay only; it does not presuppose the desired zero-free half-plane.

For any compact subset of $\operatorname{Re} s > 1 - 1/6000$, choose one positive ϵ smaller than its distance to the boundary. The two endpoint bounds, also with powers of $\log Z$, show that

$$T(s) = \int_0^\infty f_\eta(Z) Z^{-s+8/15} \frac{\mathrm{d}Z}{Z}$$

is holomorphic on that compact neighborhood. These definitions agree on overlaps because they are the same convergent integral. This is the union of half-planes argument; an endpoint-uniform big-O constant is unnecessary.

On $s = 2 + i\tau$, write $t = \log Z$ in (16). The function $\mathrm{e}^{-(2-8/15)t} f_\eta(\mathrm{e}^t)$ is integrable, and is the inverse Fourier transform of the continuous rapidly decreasing function $\mathrm{e}^{(2+i\tau-5/6)^2} H_\eta(2+i\tau)/L_F^S(2+i\tau,\eta)$. Fourier inversion therefore identifies T with this meromorphic quotient on the original line without a formal exchange of two conditionally convergent Mellin integrals. The identity theorem extends that equality to the connected larger half-plane. Its numerator is nowhere zero there, already by the source's .998 region, alternatively by Theorem 3.1. A zero of the denominator would force a pole, contradicting holomorphy of T. The finitely omitted Euler factors have zeros only on $\operatorname{Re} s = 0$, so the full Hecke function has the same exclusion. □

The improvement here is a factor 10/3 over the source's *actually proved* 1/20000, not just a comparison with its advertised 10^{-6}. It is an extraction of unused epsilon slack from an unchanged reflected moment, not a new unconditional zero-free theorem. Even the original 350000 contours suffice for Theorem 6.3; the smaller-line certificate is logically independent of it and of the shorter detector.

7 What the Arithmetic Conclusion Still Requires

The exact finite Poisson step helps locate the boundary. For allowed generators s, A of exterior integral ideals, put $g_s(k) = \sum_{d \bmod s} \chi_s(d)\mathrm{e}_S(kd/s)$ and

$$\mathcal{F}(s, A, h) = \sum_{\substack{d \bmod s \\ h \equiv Ad \ (s)}} \chi_s(d) \sum_{r \bmod A} \chi_A(r)\mathrm{e}_S((h - Ad)r/(sA)).$$

The exact finite Fourier identity is

$$\sum_{m \bmod sA} q_s^{-1/2} g_s(-m)\chi_A(m)\mathrm{e}_S(hm/(sA)) = q_s^{1/2}\mathcal{F}(s, A, h). \tag{24}$$

Expand g_s and write $m = r + Ak$, where r runs modulo A and k modulo s. The k sum is q_s exactly when $h - Ad \in sR$ and zero otherwise, by the self-dual lattice pairing. This proves the identity without assuming $(s, A) = 1$ or discarding a unit mask. The lattice index $q_s q_A$, additive dilation $q_s X$, probe factor $(q_s X)^{-1/2}$, and factor $q_s^{1/2}$ in (24) give $\sqrt{X}/q_A$. The Fourier norm is Xq_h/q_A. These normalizations explain the scale powers; they do not prove arithmetic large-sieve or Type II bounds.

Proposition 7.1 (Fixed-base reduction with two inputs)**.** *Fix an admissible integer a. Assume a fixed progression supplies $\gg_a x/(\log x)^2$ primes $p \in (x, 2x)$ with $(a/p) = -1$ and*

$$p - 1 = crQ, \quad c \in \{2, 4\}, \quad Q > x^{.9} \text{ prime}, \quad \exp((\log x)^{.1}) < q < \exp((\log x)^{.3}) \quad (q \mid r).$$

Assume also, uniformly for sufficiently large primes $q \le \exp((\log x)^{.3})$, that

$$\#\{p \in (x, 2x] : p \text{ splits completely in } \mathbb{Q}(\mu_q, a^{1/q})\} \ll_a \frac{x}{q(q-1)\log x} + x^{1-\delta_0} \tag{25}$$

for a fixed $\delta_0 > 0$, with a starting point independent of q. Then all but $o_a(x/(\log x)^2)$ of the supplied primes have $\mathrm{ord}_p(a) = p - 1$.

Proof. Let $i_p = (p-1)/\mathrm{ord}_p(a)$. The nonresidue condition makes i_p odd: otherwise the order divides $(p-1)/2$, contrary to Euler's criterion. If $i_p > 1$, some prime $q \mid i_p$ divides rQ. For $p \nmid aq$, Frobenius in the splitting field of $T^q - a$ fixes all roots precisely when $p \equiv 1 \pmod q$ and $a^{(p-1)/q} \equiv 1 \pmod p$. Writing $L = \log x$, the union of exceptions from primes dividing r therefore has size at most

$$\ll_a \frac{x}{L} \sum_{m > \exp(L^{.1})} \frac{1}{m(m-1)} + x^{1-\delta_0}\exp(L^{.3}) = o_a(x/L^2).$$

If $Q \mid i_p$, then $\mathrm{ord}_p(a) < 2x^{.1}$. Such a prime divides the nonzero integer $\prod_{1 \le j \le 2x^{.1}} |a^j - 1|$, whose logarithm is $O_a(x^{.2})$. Thus only $O_a(x^{.2}/L)$ distinct such primes exceed x. The finitely many ramified primes disappear for large x. This proves the assertion for every sufficiently large real x, with a fixed, and hence infinitude conditionally on the two inputs. □

For $a = 2$ the source uses $(M, c, u) = (8, 4, 5)$: on that progression $(2/p) = -1$. Negative squares have squarefree part -1, not 1, and remain admissible. Odd perfect powers are also allowed: the fixed prime factors of their exponent eventually lie below the factors in rQ. There is no uniform onset as the base or that exponent varies.

The splitting argument needs degree $q(q-1)$, the discriminant control $\log D_q + [K_q : \mathbb{Q}] \ll_a q(q-1)\log(2q)$, abelian factorization through $\mathbb{Q}(\mu_{12q})$, and a smooth explicit formula uniform

through degree and discriminant. These remain imported, especially across growing q. We do not turn the conditional width above into an unconditional splitting theorem. The controlled-predecessor input is equally substantive. The marked weight uses the *complete* group-prime part $h_{\mathcal{P}}$, so $F(ph) = F(h)$ must hold even when $p \mid h$. Coefficient transfer requires cancellation for every logarithmic-modulus character and $|t| \le 2XL^B$, not just bounded heights. Rough-factor densities use $\mathrm{d}y/\log y$, the factor $1/j!$, and a fixed positive exponent gap. Bombieri–Vinogradov errors, least-prime-factor subtraction and an upper bound near the square root with constants independent of the number of marks must hold before positive prime mass follows. Transforms supply none of these arithmetic estimates by themselves.

8 Comparison, Limitations, and Further Work

Primary public records were freshly consulted on October 8, 2026 using public titles and URLs only. Klurman, Shparlinski and Teräväinen [2] establish an almost-all-base result in a short growing base range; v2 is dated January 23, 2026. An exceptional set varying with x does not imply a result for every fixed admissible base. Murty and Naik's August 21, 2026 abstract [3] distinguishes an implication from a conjectural nonabelian large sieve to Artin from unconditional partial sieve results. Sgobba [4] treats number-field index sets under convergence conditions on associated series, not automatically the singleton index-one problem for a fixed rational base.

Thorner and Zaman's abstract [5] describes effective Chebotarev estimates allowing a Landau–Siegel zero, not a common fixed all-height zero-free strip. The Kazhdan–Patterson correction [6] withdraws an auxiliary assertion in the original global argument. The source's Whittaker appendix explicitly addresses it using the nonzero global functional and normalized exterior values. That distinction is retained, not silently replaced by the uncorrected product formula. Outside representation-theoretic proofs and companion arithmetic papers have not been audited in full here.

Benefits. The factorization makes phase-sensitive nonvanishing transparent, with distinct $1/2$ and $2/3$ domains. The annular construction controls both zero-frequency removal and the nonzero principal residue. The detector reduces a source length cost under explicit assumptions. The contour ledger locates the limiting bottleneck and reduces the enormous far real part. The fixed-base reduction identifies where transform work stops.

Costs and limits. The theta representation, reflection large sieve, reciprocal bounds, marked Type II estimates and prime construction remain imported. The 3481 line is optimal only within two inequalities, and its tiny slack can worsen constants. Higher weight cancellation enlarges support and seminorms; J must remain fixed. The detector has a field-dependent height range. Principal Euler nonvanishing does not widen the generic H_u regions. No result here changes L3, establishes historical priority, or certifies a global Artin or Riemann-hypothesis statement.

Feasible further work. A separate audit could verify every theta normalization against the corrected representation theory. Quantitative annular optimization could control seminorms at fixed residue normalization. Joint contour-coordinate optimization would require rechecking every Euler and reciprocal region. The marked operator and sparse-frequency estimates in companion papers need their own proof review before accepting a global conclusion. These are obligations, not promised outcomes.

9 Verification and Contribution Assessment

The independently executable standard-library script `_audit/M055-checks.py` passed 17,098 exact finite cases: 3,456 complex Euler samples; 240 annihilator-moment cases; 201 detector budgets; 5,999 integer-line feasibility cases; 10 named contour margins; 2,144 binary-prefix decompositions; and 5,048 finite index-criterion cases. Gaussian-rational pairs and rational arithmetic use no numerical acceptance tolerance. Formal disk points need not all be realized by Hecke characters; the proof, not the samples, supplies the general identity. There are no experiments claiming prime cancellation or testing the infinite-prime conclusion. Parent-run cases are separate and not included in this total.

All 16 substantive public TeX files and the public README were literally read in bounded displays: 7,468 distinct lines. The eight-line vendored font loader was inventoried and hashed but not scientifically read. The actual private original intervals were master-atoms lines 261–548 and TR1674 lines 95–321, totaling 515 distinct lines. Other private lines remain unread for this assignment. Matching pre/post hashes certify byte endpoints, not continuous monitoring or a whole-corpus audit. External literature scopes are recorded separately; most outside records were read only at abstract or metadata level.

Assessment. C3 is supported by Theorems 3.1, 4.1, 5.1, 6.3 and Propositions 6.1–6.2. The principal factor and separated domains, moment control, shorter detector and exact contour savings are notable local improvements beyond renamed coordinates. The parent suggested the limiting-width and contour choices; this report independently derives and tests them. Standard transform identities, Gram positivity and the source's existing reduction are explanatory support, not new-contribution credit. C4 is not warranted: the central global arithmetic assertions remain unverified here.

References

[1] OpenAI mathematical preprint collection, *Primitive roots for every admissible integer base*, October 4, 2026. Pinned local manuscript; public source directory. Parent handles snapshot authentication separately.

[2] O. Klurman, I. E. Shparlinski and J. Teräväinen, *On Artin's conjecture on average and short character sums*, Bull. London Math. Soc. 57 (2025), 2429–2443; arXiv:2412.13355v2, January 23, 2026. Introduction and theorem statements inspected; no full-proof audit.

[3] M. Ram Murty and S. Naik, *A non-abelian large sieve and Artin's primitive root conjecture*, arXiv:2608.21573, August 21, 2026. Primary abstract and metadata inspected.

[4] P. Sgobba, *Unconditional results for Artin-type problems over number fields*, arXiv:2508.08996, August 12, 2025. Primary abstract and metadata inspected.

[5] J. Thorner and A. Zaman, *A unified and improved Chebotarev density theorem*, Algebra Number Theory 13 (2019), 1039–1068; primary arXiv record. Abstract and metadata inspected, not the full proof.

[6] D. Kazhdan and S. J. Patterson, *Metaplectic forms*, Publ. Math. IHES 59 (1984), 35–142; correction, 62 (1985), 203. Correction prose inspected; original full representation-theoretic proof remains imported.

[7] M. Subbarao, confidential local original proofs: RamCalBooks volume 2 master-atoms chapter, Fourier/Laplace/Mellin passages; TR1674 revised-v2, arithmetic energy and exact finite Gram passages. Only the stated intervals were read. No private content was sent to public search services.